

The empirical recurrence for OEIS A204699, proved by transfer matrix

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Abstract

OEIS A204699 counts arrays of width 2 over $\{0, 1, 2\}$ containing no run of 3 consecutive cells, taken along a named direction of the grid, whose entries are all equal. The entry carries an empirical recurrence of order 3 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: every such run lies inside 3 consecutive array rows, so the count is a walk count on $S = 115$ states.

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1 The conjecture

OEIS A204699 is “Number of $n \times 2$ arrays with no occurrence of three equal elements in a row horizontally, vertically or nw-to-se diagonally, and new values introduced in row major order.”. It has offset 1 and begins

2, 14, 96, 726, 5400, 40344, 301056, 2247264, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 6*a(n-1) + 12*a(n-2) - 8*a(n-3)$ for $n > 5$.

As of the “Last modified” line on the live entry (Feb 23 2018 09:51:35, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

A RUN is 3 consecutive cells taken along one of the directions

along a row
down a column
down a diagonal, nw to se

all of them inside the array. The array is counted when no run has its entries all equal.

The entry closes with “new values introduced in row major order”, so what is counted is arrays up to renaming the letters. That is legitimate here because the forbidden runs are the runs of EQUAL entries, and whether entries are equal is untouched by any renaming. Writing N_j for the classes on exactly j letters and L_i for the admissible arrays over an alphabet of i letters,

$$L_i = \sum_j N_j i(i-1) \cdots (i-j+1), \quad a = N_1 + \cdots + N_3 = (\mathbf{1}^\top F^{-1})L.$$

The neighbouring entries that forbid INCREASING runs carry no such clause, and could not: a permutation of the alphabet does not preserve order.

3 Every run lies in 3 consecutive rows

Lemma 1. *Each direction above changes the row index by 0 or 1 from one cell of a run to the next, so a run of 3 cells lies inside 3 consecutive array rows, and a run whose last cell is in row i lies inside rows $i - 2, \dots, i$.*

Proof. Immediate from the row steps of the four directions, which are 0 for a row and 1 for the other three. $\square$

So the state is the window of the last 2 array rows, a row not yet written carried as $\perp$; a step appends a row and rejects it exactly when some run ends in it or lies inside it. A run needing a row that is $\perp$ does not exist, which is the boundary rule and is read off the window rather than assumed. The walk starts at the all- $\perp$ window, so a walk of R steps is an array of R rows. Assembling the pieces for $i = 1, \dots, 3$ as a disjoint union, with ι carrying those weights cleared of denominators,

$$a(n) = \iota^\top M^j \tau, \quad S = 115, \quad \tau = \mathbf{1}.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 2. *Let $q(t) = t^3 - \sum_i c_i t^{3-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+3} - \sum_i c_i M^{j+3-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) = 6a(n-1) + 12a(n-2) - 8a(n-3)$$

for every $n > 5$.

Proof. The vector $w = q(M) \tau$ was formed by 3 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 5 + 1$ onwards and the run continued until the working vector was identically zero. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells in row major order, in the orientation the entry's own name writes, and reject as soon as a completed run is all equal. No window, no transposition and no transfer matrix are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.